

SMART INDIA HACKATHON 2026



NAWI-ReportPro: OIML R-76 Digital Platform

Problem Statement ID: 26035

Problem Statement Title:

Automated Test Report Generator for
Non-Automatic Weighing Instruments (NAWI)

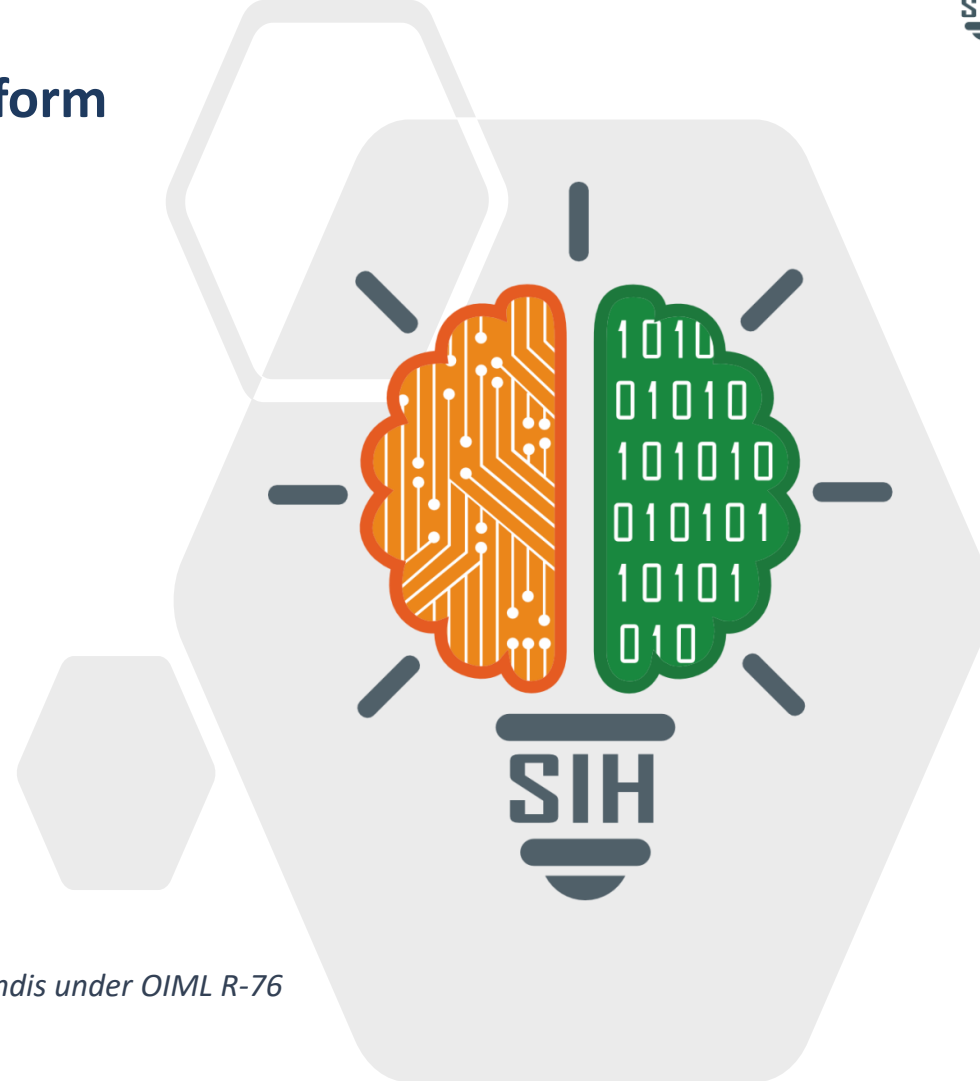
Theme: Smart Automation

PS Category: Software

Team ID: SIH2026-TEAM-NAWI

Team Name: Metrology Innovators

Institution: [Your College Name Here] | Mentor: [Faculty SPOC Name]



Automating India's Legal Metrology Verification across 1,656 Agricultural Mandis under OIML R-76

IDEA TITLE: NAWI-ReportPro — Automated Digital Metrology

Proposed Solution & Core Innovations

- **Digital Transformation:** Replaces manual paper verification for Non-Automatic Weighing Instruments (scales & truck weighbridges) with an end-to-end digital system.
- **Direct Scale Telemetry:** Connects directly to weighing indicators via browser Web Serial API (RS-232/USB) — live weight capture with zero manual typing errors.
- **OIML R-76 Engine:** Automates Maximum Permissible Error (MPE) calculations across all 4 accuracy classes (Class I to IIII) with multi-interval tare compensation.
- **Tamper-Proof Certificates:** Issues legal verification certificates secured with HMAC-SHA256 digital seals and instant QR code public verification.

KEY METROLOGICAL & ECONOMIC INNOVATION

Direct browser-to-indicator Web Serial API eliminates ₹50,000 proprietary hardware converters per mandi — runs on existing mandi weighbridge PCs with zero driver installation.

① OIML R-76 Compliant Engine

Automated MPE calculations for Class I–III
Multi-interval tare compensation • Tolerance validation

② Live RS-232 Telemetry

Real-time weight capture from Mettler Toledo SICS,
Avery Weigh-Tronix & Essae indicators • Zero typing errors

③ Tamper-Proof Digital Certificates

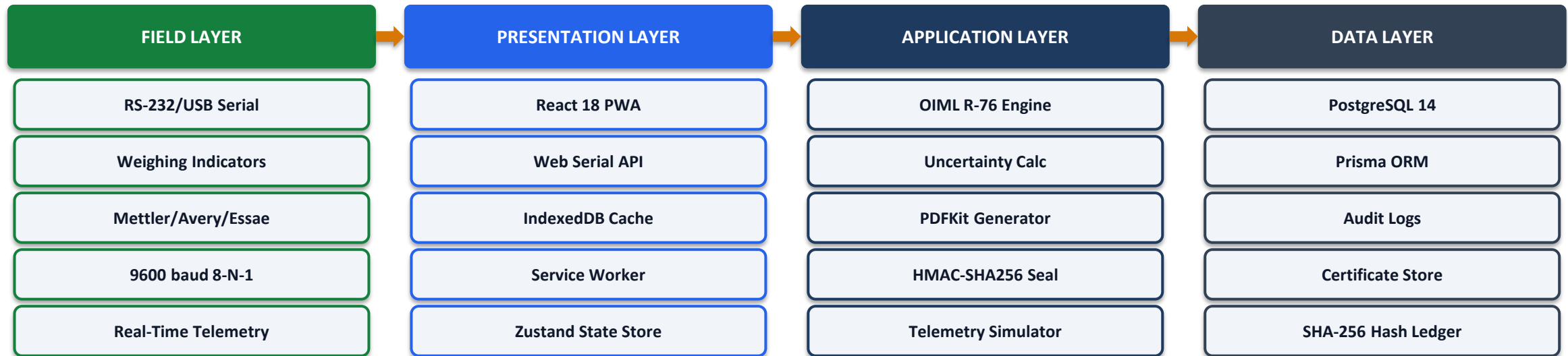
HMAC-SHA256 cryptographic seals • QR verification
Public portal at /verify/:certificateNo • Immutable audit trail

④ Offline-First PWA

IndexedDB local cache • Background auto-sync
Works in zero-connectivity rural mandis • Zero data loss

TECHNICAL APPROACH: System Architecture & Stack

- **Hardware Layer:** Browser Web Serial API reads continuous ASCII/HEX telemetry (9600 baud, 8-N-1) from Mettler Toledo SICS, Avery Weigh-Tronix & Essae indicators.
- **Application Stack:** React 18 PWA + Node.js Express REST API + PostgreSQL 14 (Prisma ORM) + PDFKit certificate engine + HMAC-SHA256 signer.
- **Metrology & Offline:** OIML R-76 MPE engine (all 4 accuracy classes), ISO GUM uncertainty budget ($k=2$), IndexedDB queue with idempotent background sync.



System Performance: <15ms OIML MPE computation | 100% Offline-capable PWA | Zero vendor-lock RS-232 parser | 129/129 tests passing

✓

Working Prototype: 15 core features built & validated | 129/129 automated tests passing (100%) | Zero licensing cost (FOSS stack)

Feasibility Analysis ✓	Challenges & Risks ⚠	Mitigation Strategies 🛡
Technical: Sub-15ms OIML MPE computation across all 4 accuracy classes; ISO GUM uncertainty budget (k=2)	Legacy indicator protocol variance: Non-standard RS-232 ASCII formats across 1,656 mandis	Protocol-agnostic regex parser with auto-baud detection (9600/4800) + virtual simulator fallback for field training
Operational: Offline PWA runs on any ₹10K Android tablet or refurbished laptop via Chrome browser	Zero 4G/WiFi coverage in remote agricultural mandi yards and rural warehouses	IndexedDB offline transaction log with idempotent background auto-sync + SHA-256 conflict resolution on reconnect
Economic: Zero licensing fees (FOSS stack); no proprietary hardware adapters needed	Inspector resistance: Field officers hesitant to adopt digital certificates over familiar paper books	3-step guided workflow reducing test entry to <15 min + regional language UI (Hindi/Telugu/Marathi) + QR spot-checks
Regulatory: Conforms to OIML R-76 (2006/E) & Legal Metrology Rules 2011 (incl. July 2026 amendments)	Certificate tampering and forgery risk for printed inspection slips at mandi gates	HMAC-SHA256 cryptographic seal in offline-generated PDF + instant public QR verification at /verify/:certNo

IMPACT AND BENEFITS: National Scale Outcomes

- **Economic Savings:** Eliminates ₹420+ Crores annual procurement weighing discrepancies; reduces inspector labor cost by 87% (from 2 hrs to <15 min per instrument).
- **Farmer Fraud Protection:** Guarantees fair pricing for 1.80 Crore farmers across 1,656 e-NAM mandis; transparent dispute resolution via public QR scan of legal calibration seal.
- **Governance & Environmental:** Eliminates 50+ lakh paper verification certificates annually; provides DoCA real-time national compliance heatmaps across all State Directorates.
- **Inspector Throughput:** Increases daily inspector capacity from 4 weighbridges/day (manual) to 25+ weighbridges/day (digital), eliminating the national verification backlog.

10x

FASTER

Field Verification
Cycle Time

1,656

MANDIS

Agricultural Markets
Covered Nationwide

1.80 Cr

FARMERS

Protected from
Weight Manipulation

100%

TAMPER-PROOF

HMAC-SHA256
Digital Seal Security

MANUAL: 2 hours per instrument >>> DIGITAL: Under 15 minutes | 87.5% efficiency gain

1. International Metrology Standards

- **OIML R 76-1:2006:** Non-automatic weighing instruments — Metrological & technical requirements; Table 3 MPE limits.
- **OIML R 76-2:2007:** Standardized test report format and repeatable testing procedures.
- **ISO/IEC Guide 98-3 (GUM):** Expanded uncertainty budget calculation ($k=2$, 95.45% confidence level).
- **EURAMET cg-18 v4.0:** Calibration guidelines — repeatability, eccentricity & temperature formulas.
- **WELMEC Guide 7.2:** Software Guide (Measuring Instruments Directive 2014/32/EU) — Risk Class D compliance.

2. Indian Legal Framework & Government Portals

- **Legal Metrology Act, 2009:** Statutory mandates for verification & calibration of commercial weighing instruments.
- **Legal Metrology Rules, 2011:** Seventh Schedule standards, incl. July 2026 amendments for high-capacity scales (≥ 1 tonne).
- **BIS IS 9281 (Parts 1-4):** Bureau of Indian Standards specs for electronic weighing systems and load cells.
- **DoCA eMaap Portal:** Department of Consumer Affairs online enforcement and inspection tracking system.
- **e-NAM Platform:** National Agriculture Market: 1,656 mandis, 1.80 Cr farmers (June 2026 official data).
- **CSIR-NPL:** National Physical Laboratory — apex metrological standards & calibration traceability.

Feature / Metric	NAWI-ReportPro	Manual Paper	DoCA eMaap	Proprietary SCADA
Direct RS-232 Telemetry	✓ Zero typing	✗ Manual entry	✗ Manual form	✓ Vendor-locked
OIML R-76 MPE Engine	✓ Instant (all classes)	✗ Manual calc	✗ No validation	⚠ Basic tolerance
Offline PWA (Remote Mandis)	✓ IndexedDB + sync	✓ Paper-based	✗ Needs internet	✗ On-prem server
Tamper-Proof Certificates	✓ HMAC-SHA256 + QR	✗ Forgeable stamps	⚠ Basic DB entry	✗ Proprietary logs
Hardware Independence	✓ Any scale vendor	N/A	N/A	✗ OEM only
Total Software Cost	₹0 (FOSS Stack)	High recurring	Govt funded	₹5L–15L / site